

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Electronics And Communication Engineering Department

Microwave Engineering Equipment

S. No.	Major Equipment	Purpose / Application
1	Microwave Test Bench (X-Band)	Microwave experiments and waveguide measurements
2	Antenna Trainer Kit (SICO DSR-06(P))	Study of antenna characteristics and radiation patterns
3	Antenna Trainer Kit (SCIENTECH ST2261)	Antenna experiments and measurements
4	Antenna Trainer Kit (SCIENTECH ST2263)	Advanced antenna studies
5	Motorized Antenna Positioner with Software (HOLMARC)	Automated antenna radiation pattern measurement
6	Vector Network Analyzer (VNA) with Spectrum Analyzer (ANRITSU MS2037C)	S-Parameter measurement, antenna and filter characterization
7	Signal Generator (Rohde & Schwarz SMC100A)	RF and microwave signal generation
8	Microwave Power Meter with Sensor (SICO)	Microwave power measurement
9	Klystron Power Supply Unit	Power supply for Reflex Klystron experiments
10	Reflex Klystron Mount with Tube	Microwave source for X-band experiments
11	Gunn Power Supply Unit	Power supply for Gunn Oscillator
12	Gunn Oscillator	Microwave signal generation
13	VSWR Meter	Standing Wave Ratio measurement
14	Frequency Meter (Direct Type)	Frequency measurement
15	Frequency Meter (Micrometer Type)	Precision microwave frequency measurement
16	Waveguide Slotted Line Section	Measurement of standing waves and impedance
17	Tunable Detector Mount	Microwave signal detection
18	Tunable Probe	Electric field measurement in waveguides
19	PIN Modulator	Microwave modulation experiments
20	Variable Attenuator	Variable attenuation of microwave signals
21	Fixed Attenuators (3 dB, 6 dB, 10 dB)	Signal attenuation
22	Multi-Hole Directional Coupler	Coupling coefficient, directivity and insertion loss measurement
23	Cross Directional Coupler	Microwave signal coupling measurements
24	Magic Tee	Hybrid junction experiments

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25	E-Plane Tee	Power division and microwave network experiments
26	H-Plane Tee	Power division and microwave network experiments
27	Three-Port Ferrite Circulator	Signal routing and isolation experiments
28	Isolator	Protection against reflected microwave power
29	Slide Screw Tuner / Precision Slide Screw Tuner	Impedance matching
30	E-H Tuner / Precision E-H Tuner	Microwave impedance matching
31	Precision Sliding Termination	Reflection coefficient measurements
32	Matched Termination Load	Waveguide termination
33	Waveguide Filter	Microwave filtering experiments
34	Coaxial-to-Waveguide Adapter	Transition between coaxial and waveguide systems
35	Horn Antenna	Antenna gain and radiation experiments
36	X-Band Waveguide Short Section	Waveguide propagation experiments
37	E-Plane Bend	Waveguide transmission experiments
38	H-Plane Bend	Waveguide transmission experiments
39	Liquid Dielectric Cell	Dielectric constant measurements
40	Transmission Line Trainer Kit (SCIEN TECH ST2266)	Transmission line experiments
41	DC Power Supply (Keithley 2231A-30-3)	Power source for laboratory experiments
42	Desktop Computers	Data acquisition, simulation and instrument control
43	CRO (Sciencetech CADDO 805)	Signal waveform observation
44	Microwave Components Kit (LPF, VCO, Mixer, LNA, Multiplier/Amplifier)	RF and Microwave circuit experiments

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Figure 1. Microwave bench for X band measurement and analysis



Figure 2. Antenna trainer kit radiation pattern measurement of different type of antennas